

Aliyeva Khadija (2026). Production of Ethylbenzene by Liquid-Phase Alkylation of Benzene and Ethylene over Zeolite Catalyst: Process Technology and Material Balance. Literature-Based Project.

Unpublished Project, Baku.

1.Introduction to Ethylbenzene Production

Ethylbenzene is considered one of the important aromatic compounds in the petrochemical industry. Ethylbenzene is highly flammable and is a liquid with a gasoline-like odor. 99% of ethylbenzene produced on an industrial scale is used to produce styrene. Styrene is the main raw material for the production of polystyrene, styrene-butadiene rubber, acrylonitrile butadiene styrene, etc. The remaining small part is used in various fields:

As a solvent in paints, varnishes, rubber and chemical industries

As an anti-detonation additive to increase the octane number of benzene

As an intermediate in the production of products such as acetophenone and ethylbenzene peroxide

It can be injected underground during natural gas production.



Ethylbenzene is produced in industry by alkylation of benzene with ethylene in the presence of a catalyst. Although AlCl_3 has traditionally been used as a catalyst, a switch to zeolite-type catalysts has been made due to its many advantages. Increasing the selectivity of this transition process has created conditions for reducing problems related

to corrosion and waste. As zeolite-type catalysts, catalysts such as ZSM-5 Y-zeolite Zeolite Beta MCM-22 can be used during the process.

During the production of ethylbenzene, diethylbenzene is obtained as a by-product. This can lead to problems such as the purification of the final product due to the decrease in the efficiency of ethylbenzene production.

The diethylbenzene formed as a result of the reaction is returned to the process by transalkylation. In this case, diethylbenzene reacts with benzene to obtain ethylbenzene.

The alkylation process can produce unreacted benzene, diethylbenzene, ethylene, and other by-products in addition to the intended product ethylbenzene. In this regard, the separation process is very important. Unreacted benzene can be recycled, and diethylbenzene and other heavy components can be returned to the process by transalkylation.

2.Process Conditions

The alkylation process can be carried out in both liquid and gas phases. If the process is carried out in the gas phase, high temperatures are required, while in the liquid phase, relatively lower temperatures are required. This project will discuss the production process of ethylbenzene in the liquid phase with the participation of a zeolite catalyst.

During the alkylation process, benzene and ethylene interact on the surface of a zeolite-based catalyst. The catalyst composition is approximately 90% Y-zeolite and 10% alumina (Al_2O_3).

The reactor temperature is assumed to be 245-270 C and the pressure is assumed to be 500 psig. The molar ratio of benzene to ethylene is 7.2:1.

Polyethylbenzene (PEB) formed during alkylation is sent to the separation stage from the main product, which mainly includes diethylbenzene and triethylbenzene. After separation, ethylbenzene is produced by the interaction of the PEB fraction with benzene in a fixed bed transalkylation reactor.

The transalkylation reactor operates at a temperature of approximately 250 C and a pressure of 510 psig. The molar ratio of benzene to PEB is assumed to be 13:1. During the process under these conditions, the PEB conversion is approximately 92.9% and the selectivity for ethylbenzene is determined to be 100%.

	A	B	C
1	R-101 / R-102 — Reactor Specifications (SRI/PEP Datasheet)		
2			
3	Parameters	R-101 Alkylation Reactor	R-102 Transalkylation Reactor
4	Reactor type	Fixed bed, four stages with interstage cooling	One fixed bed
5	Catalyst	Y-zeolite (90%) + alumina hydrogel (10%)	Y-zeolite (90%) + alumina hydrogel (10%)
6	Catalyst life (years)	2	2
7	Reactor temperature (°C)	245-270	250
8	Reactor pressure (psig)	500	510
9	WHSV, total feed (lb feed/lb cat/h)	5,9	6
10	WHSV, ethylene (lb C ₂ H ₄ /lb cat/h)	0,27	--
11	Feed ratio (mol/mol)	7.2 (benzene/ethylene); 6.8 (phenyl/ethyl)	13 (benzene/PEB); 5.8 (phenyl/ethyl)
12	Conversion (%)	99.5 on ethylene	13.4 on benzene; 92.9 on PEB
13	Selectivity to EB (mol%)	91,6	~100
14	Selectivity to PEB (mol%)	3,3	--
15			
16	Overall Selectivity to EB (mol%)	98,2	
17	Plant Yield on Ethylene (mol%)	98,2	
18	Plant Yield on Benzene (mol%)	98,4	
19			
20	Components		
21	Ethylene	raw material	39.285 lb/hr
22	Benzene	raw material	109.054 lb/hr
23	EB	product	145.710 lb/hr
24	Xylene	by-product	-
25	P-EB	by-product/ Intermediate product	-
26	H ₂ O	Contituent of raw material	-
27	Heavies	by-product	-
28			

3. Material balance equation for R-101 reactor

$$1. \bar{m}_{\text{ethylene}} = f_{\text{net}} * \bar{m}_{EB} = 0.26913 * 145710 = 39285 \text{ lb/h}$$

$$\bar{m}_{\text{benzene}} = f_{\text{net}} * \bar{m}_{EB} = 0.74844 * 145710 = 109055 \text{ lb/h}$$

2. Conversion of mass flows to molar flow

$$n = \bar{m} / M$$

$$\text{Benzene (C}_6\text{H}_6) \ n = 109.054 / 78.11 = 1396,1 \text{ lbmol/h}$$

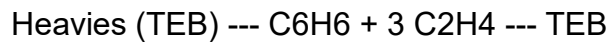
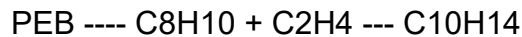
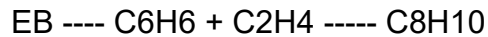
$$\text{Ethylene (C}_2\text{H}_4) \ n = 39.285 / 28.05 = 1400,5 \text{ lbmol/h}$$

Molar amount of ethylene entering and not entering the reaction:

$$(\text{C}_2\text{H}_4) \ n_{\text{reacted}} = 0.995 * 1400.5 = 1393.5 \text{ lbmol/h}$$

$$(\text{C}_2\text{H}_4) \ n_{\text{unreacted}} = 0.005 * 1400.5 = 7 \text{ lbmol/h} = 196 \text{ lb/h}$$

3. Stoichiometric distribution (on C₂H₄)



$$n_{\text{ethylene path}} = S_{EB} * n_{\text{reacted}}$$

$$n_{\text{ethylene path EB}} = 0.916 * 1393,5 = 1276,5 \text{ lbmol/h} \quad \text{C}_2\text{H}_4 / \text{EB} = 1/1$$

$$n_{\text{ethylene path PEB}} = 0.033 * 1393,5 = 46 \text{ lbmol/h} \quad \text{C}_2\text{H}_4 / \text{PEB} = 2/1$$

$$n_{\text{ethylene path TEB}} = 0.051 * 1393,5 = 71,1 \text{ lbmol/h} \quad \text{C}_2\text{H}_4 / \text{TEB} = 3/1$$

Taking into account stoichiometry:

$$n_{EB} = 1276,5 \text{ lbmol/h}$$

$$n_{PEB} = 46/2 = 23 \text{ lbmol/h}$$

$$n_{TEB} = 71,1/3 = 23,7 \text{ lbmol/h}$$

Expressing in mass

$$\bar{m} = n * MW$$

$$\bar{m}_{EB} = 1276,5 * 106,17 = 135517 \text{ lb/h}$$

$$\bar{m}_{PEB} = 23 * 134,22 = 3087 \text{ lb/h}$$

$$\bar{m}_{TEB} = 23,7 * 162,27 = 386 \text{ lb/h}$$

4. Calculation relative to benzene

Benzene entering the reactor

$$n_{\text{benzene}} = n_{\text{fresh}} + n_{\text{recycle}}$$

$$n_{\text{Benzene}} = 7.2 * 1400,5 = 10083,6 \text{ lbmol/h} = 787631 \text{ lb/h}$$

$$n_{\text{Benzene reacted}} = 1276,5 + 23 + 23,7 = 1323,2 \text{ lbmol/h}$$

$$n_{\text{Benzene recycle}} = 10083,6 - 1323,2 = 8760,4 \text{ lbmol/h} = 684,300 \text{ lb/h}$$

5. Application of mass balance

$$\Sigma \bar{m}_{\text{in}} = \Sigma \bar{m}_{\text{out}}$$

$$\bar{m}_{\text{benzene}} + \bar{m}_{\text{ethylene}} = \bar{m}_{\text{benzene recycle}} + \bar{m}_{\text{EB}} + \bar{m}_{\text{PEB}} + \bar{m}_{\text{TEB}} + \bar{m}_{\text{unreacted}}$$

$$787631 + 39285 = 684,300 + 135517 + 3087 + 3846 + 196$$

Material balance equation for R102 transalkylation reactor

1. Amount of PEB moles from R-101:

$$n_{\text{PEB in}} = 23 \text{ lbmol/h}$$

$$\bar{m}_{\text{PEB in}} = 23 * 13.22 = 3087 \text{ lb/h}$$

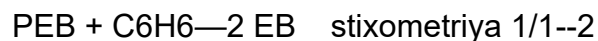
2. Calculation based on benzene

$$\text{Feed ratio benzene/PEB} = 13 / 1$$

$$n_{\text{benzene}} = 13 * n_{\text{PEB in}} = 13 * 23 = 299 \text{ lbmol/h}$$

$$\bar{m}_{\text{benzene in}} = 299 * 78,11 = 23355 \text{ lb/h}$$

3. Calculation based on reaction



$$n_{\text{PEB reacted}} = n_{\text{PEB in}} * X_{\text{PEB}} = 0.929 * 23 = 21,4 \text{ lbmol/h}$$

$$n_{\text{benzene reacted}} = 21,4 \text{ lbmol/h}$$

$$n_{\text{EB}} = 2 * 21,4 = 42,7 \text{ lbmol/h}$$

$$\bar{m}_{\text{EB}} = 42,7 * 106,17 = 4534 \text{ lb/h}$$

4. Calculation of the amount of non-reacted substances:

$$n_{\text{PEB unreacted}} = 23 - 21,4 = 1,6 \text{ lbmol/h}$$

$$\bar{m}PEB \text{ unreacted} = 1.6 * 134.22 = 215 \text{ lb/h}$$

$$n \text{ benzene unreacted} = 299 - 21.4 = 277.6 \text{ lbmol/h}$$

$$\bar{m}benzene \text{ unreacted} = 277.6 * 78.11 = 21687 \text{ lb/h}$$

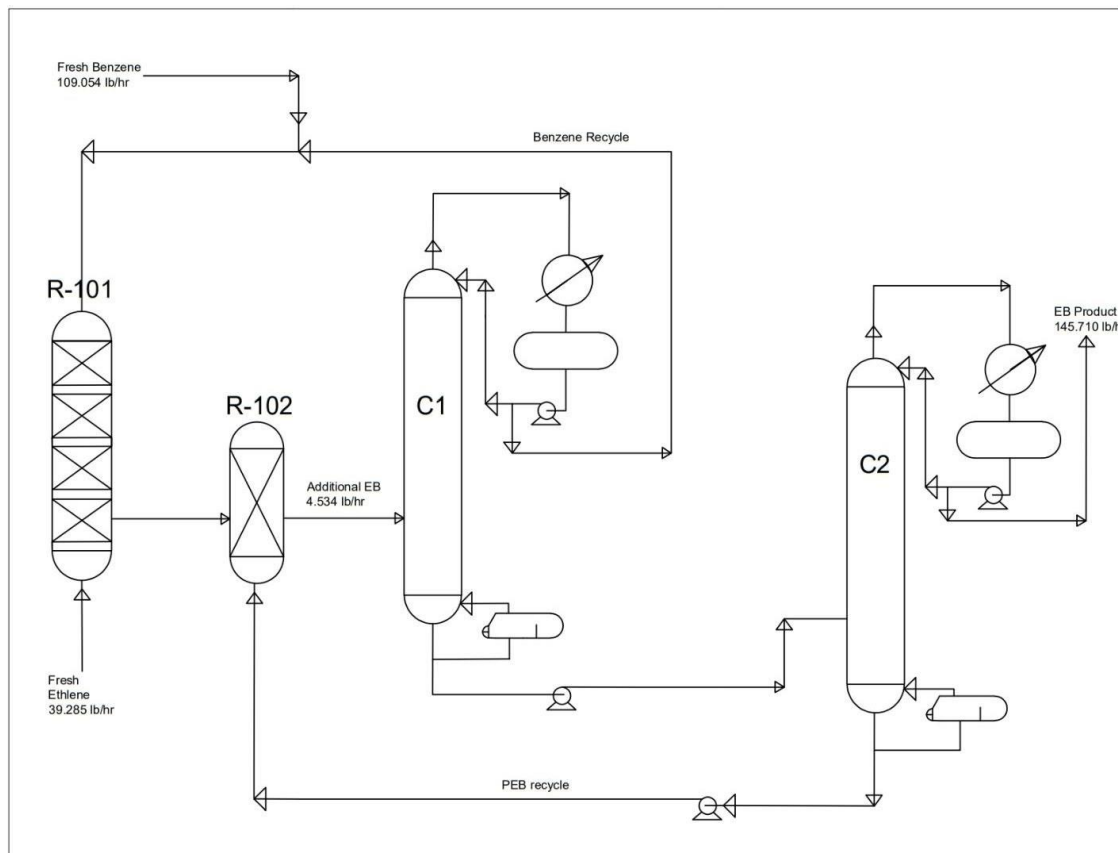
5. Application of mass balance

$$\Sigma \bar{m}_{in} = \Sigma \bar{m}_{out}$$

$$\bar{m}PEB \text{ in} + \bar{m} \text{ benzene in} = \bar{m}EB + \bar{m}PEB \text{ unreacted} + \bar{m}benzene \text{ unreacted}$$

$$3087 + 23355 = 4534 + 215 + 21687$$

4. Process Flow Diagram (PFD) of Ethylbenzene Production



The main equipment of the ethylbenzene production process:

- R-101 Alkylation reactor
A four-stage (catalyst layered) fixed-bed reactor with cooling between stages. Benzene and ethylene react over a Y-zeolite + alumina hydrogel catalyst at 245-270°C and 500 psig to form ethylbenzene.

- **R-102 Transalkylation reactor**
A single-stage (catalyst layered) fixed-bed reactor operating at 250°C and 510 psig. Ethylbenzene is formed by the reaction of diethylenebenzene (PEB) formed in R-101 with benzene.
- **C1 Distillation column**
Separates unreacted benzene from light components.
A condenser at the top of the column converts the vapor phase into a liquid phase, returns a portion to the column as reflux, and the remaining portion is returned to the reactors as benzene recycle.
The reboiler in the lower part of the column partially evaporates the heavy components and returns some of them to the column, while the rest is transferred to the C2 column.
- **C2 Distillation Column**
Separates Ethylbenzene from PEB
The condenser in the upper part of the column separates the final Ethylbenzene product as a distillate.
The reboiler in the lower part of the column separates the PEB for return to the transalkylation reactor (R-102).

5.Assumptions and Discussion:

I checked the material balance using 2 different methods.

First, when I calculated using the net consumption factors (0.26913 lb of ethylene and 0.74844 lb of benzene for every 1 lb of EthylBenzene consumed), the result was 145,710 lb/h of EthylBenzene produced, which is consistent with the figures in the SRI datasheet.

Secondly, I calculated using the R-101 reactor's own selectivity indicators. This time the result was not what I expected, 135,517 lb/h, about 10% lower than the actual indicator. At first, I thought there was an error in the datasheet, but since the source I was using was an already approved patent, this possibility was very low. After some research, it turned out that the problem was not in the calculation. The selectivity indicator I used was related to the flow with recirculation inside the reactor. Part of the PEB formed from the first reactor, R-101, is converted into EthylBenzene by reacting with R-102 benzene in the second reactor. When this is taken into account, 140,051 lb/h is obtained, but this is 4% below the target. I attributed the remaining difference to the fact that the exact composition of the heavy by-products is not given in the datasheet.

For all these reasons, I included only the first method as the result in my report, since it fully corresponds to the official values. The second method was very useful for me to understand the process more deeply.

6. References

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